

Pricing and hedging of financial derivatives
Written exam of January 29, 2021
No documents allowed

Exercise 1: gamma swap and implied volatility

A gamma swap is a modification of the variance swap where the variance of the underlying is weighted by the underlying's value. The pay-off (using the continuous approximation) is thus given by

$$H_T = \frac{1}{T} \int_0^T \frac{S_t}{S_0} \sigma_t^2 dt - \sigma_K^2,$$

where σ_K is the contractual volatility. We shall call the part

$$\frac{1}{T} \int_0^T \frac{S_t}{S_0} \sigma_t^2 dt$$

the variable leg of the gamma swap. In this exercise we suppose that the **interest rate is zero** and the underlying asset S follows the dynamics

$$\frac{dS_t}{S_t} = \mu_t dt + \sigma_t dW_t$$

under the real-world probability $\mathbb{P}$, where μ and σ are stochastic processes. Finally, we assume that call and put options of all strikes with maturity T are traded in the market, and their prices at time $t = 0$ are denoted by $C(T, K)$ and $P(T, K)$.

1. Applying the Itô formula to $f(S_t)$, show that it is possible, using an appropriate function f , to represent the variable leg of the gamma swap as a combination of a function of S_T , denoted by f_G and a stochastic integral with respect to S_t . Show that it is possible to choose the function f_G such that $f_G(S_0) = f'_G(S_0) = 0$. In the rest of the exercise we shall assume that such a choice has been made.

For this question, recall that

$$\int \log x \, dx = x \log x - x + \text{const.}$$

2. Deduce a method for replicating the gamma swap using a static portfolio of European options and a dynamic trading strategy in the underlying. Compute the fair value of σ_K .
3. Plot the function f_G on a graph. Justify that it is not possible to super-replicate the gamma swap with a portfolio containing a finite number of call and put options. Is the super-replication possible with a portfolio containing a finite number of calls and puts plus one option with quadratic pay-off ($H_T = S_T^2$)?
4. Let $C^{BS}(T, K, \sigma)$ and $P^{BS}(T, K, \sigma)$ be the Black-Scholes call and put option prices at time $t = 0$ with volatility σ . Show that

$$C^{BS}(T, K, \sigma) = \frac{K}{S_0} P^{BS}\left(T, \frac{S_0^2}{K}, \sigma\right).$$

5. Assume that the market implied volatility $I(T, K)$ has the following symmetry property $I(T, K) = I(T, S_0^2/K)$ (in other words, the implied volatility is symmetric in log-moneyness). Show that in this case the call and put option prices satisfy:

$$C(T, K) = \frac{K}{S_0} P\left(T, \frac{S_0^2}{K}\right).$$

6. Show that under the condition of the previous question, the gamma swap has the same value as the variance swap (that is, the fair value of σ_K is the same). Recall that for the variance swap, in the absence of interest rate, the fair value of σ_K is given by

$$\sigma_K^2 = \frac{2}{T} \int_0^{S_0} \frac{P(T, K)}{K^2} dK + \frac{2}{T} \int_{S_0}^{\infty} \frac{C(T, K)}{K^2} dK.$$

Exercise 2: portfolio optimization and indifference pricing

We consider a financial market with a risk-free asset, whose interest rate is constant and equal to r and a risky asset with dynamics

$$dS_t = rS_t dt + \sigma S_t (\lambda dt + dW_t).$$

The parameters λ and σ are constant.

1. Write down the dynamics of a self-financing portfolio with initial value x , which is managed by investing the *amount* π_t in the risky asset at any date t . The value of this portfolio at date t will be denoted $X_t^{x, \pi}$. Write down the discounted portfolio value $\tilde{X}_t^{x, \pi}$ as function of the discounted amount $\tilde{\pi}_t$.

Part 1: portfolio optimization We consider an economic agent with utility function $U(x) = -e^{-\gamma x}$ for $\gamma > 0$. This agent seeks to solve the following portfolio optimization problem.

$$\max_{\pi \in \mathcal{A}} \mathbb{E}[U(\tilde{X}_T^{x, \pi})], \quad (1)$$

where $\mathcal{A}$ is the set of eligible strategies. As in the course, a portfolio is said to be admissible if there exists a continuous $\mathbb{Q}$ -martingale M such that $\tilde{X}_t^{x, \pi} \geq M_t$ almost surely for all $t \in [0, T]$, where $\mathbb{Q}$ is the risk-neutral probability.

2. Let Z_T be the density of the risk-neutral probability measure. Determine the explicit form of Z_T then show that any admissible portfolio satisfies

$$\mathbb{E}[Z_T \tilde{X}_T^{x, \pi}] \leq x.$$

3. Let us consider the Fenchel transform of U :

$$V(y) = \sup_x \{U(x) - xy\}.$$

Using question 2, show that for any admissible strategy π and for all $y \geq 0$,

$$\mathbb{E}[U(\tilde{X}_T^{x, \pi})] \leq \mathbb{E}[V(yZ_T)] + xy.$$

4. Show that the Fenchel transform of the utility function $U(x) = -e^{-\gamma x}$ is given by

$$V(y) = \frac{y}{\gamma} \log \frac{y}{\gamma} - \frac{y}{\gamma}, \quad y > 0.$$

By minimizing over y , deduce from the previous question that

$$\mathbb{E}[U(\tilde{X}_T^{x,\pi})] \leq -e^{-\gamma x - \frac{\lambda^2 T}{2}}.$$

5. For this question only, assume that $\lambda = 0$. Show that in this case the optimal strategy is not to invest in stock. Provide an intuition for this result.
6. Compute explicitly the value $\tilde{X}_T^{x,\pi}$ when the discounted amount $\tilde{\pi}_t$ is constant: $\tilde{\pi}_t = \tilde{\pi}$, and the corresponding expected utility value.
7. Conclude by determining the optimal investment strategy.

Part 2: indifference pricing We now consider a bounded contingent claim G and study the portfolio optimization problem for an agent who has purchased the contingent claim G for the price $p(G)$:

$$\max_{\pi \in \mathcal{A}} \mathbb{E}[U(\tilde{X}_T^{x-p(G),\pi} + \tilde{G})]. \quad (2)$$

The buyer's indifference price of the claim G is the value $p(G)$ such that the agent is indifferent between purchasing and not purchasing the claim:

$$\max_{\pi \in \mathcal{A}} \mathbb{E}[U(\tilde{X}_T^{x-p(G),\pi} + \tilde{G})] = \max_{\pi \in \mathcal{A}} \mathbb{E}[U(\tilde{X}_T^{x,\pi})].$$

Similarly, one can define the seller's indifference price $p'(G) = -p(-G)$. In questions 8–10 we do not assume that the claim G may be replicated, it may for example depend on factors other than the Brownian motion W .

8. Give an explicit formula for the buyer's indifference price in terms of the expressions (1) and (2) and show that $p(G)$ does not depend on the initial wealth x of the agent.
9. Prove a simplified formula for the buyer's indifference price:

$$p(G) = -\frac{1}{\gamma} \log \min_{\pi \in \mathcal{A}} \mathbb{E}^{\mathbb{Q}}[e^{-\gamma(\tilde{X}_T^{0,\pi} + \tilde{G})}],$$

where $\mathbb{Q}$ is the risk-neutral probability. Give the similar formula for the seller's indifference price.

10. Using the previous question, prove the following properties of the indifference price: monotonicity (if $G_1 \geq G_2$ a.s., then $p(G_1) \geq p(G_2)$), convexity ($p(\frac{1}{2}(G_1 + G_2)) \leq \frac{1}{2}(p(G_1) + p(G_2))$), relationship between buyer's and seller's price: $p(G) \leq p'(G)$. Hint: to show convexity, use the Cauchy-Schwarz inequality.
11. In the last question we assume that the claim G admits a replicating strategy π_G . Show that in this case, $p(G) = \mathbb{E}^{\mathbb{Q}}[\tilde{G}]$.

Solution

Exercise 1

1. Itô's formula yields:

$$f(S_T) = f(S_0) + \int_0^T f'(S_t) dS_t + \frac{1}{2} \int_0^T f''(S_t) S_t^2 \sigma_t^2 dt.$$

Taking f_G such that $f_G''(x) = \frac{2}{TxS_0}$ we therefore have

$$\frac{1}{T} \int_0^T \frac{S_t}{S_0} \sigma_t^2 dt = f(S_T) - f(S_0) - \int_0^T f'(S_t) dS_t$$

Direct integration gives:

$$f'_G(x) = \frac{2}{TS_0} \log \frac{x}{S_0},$$

where the constant has been chosen to respect the condition $f'(S_0) = 0$. Another integration gives:

$$f_G(x) = \frac{2}{TS_0} (x \log \frac{x}{S_0} - x + S_0),$$

where once again we chose the constant to have $f_G(S_0) = 0$.

2. Using the formula from the course, we have:

$$f_G(x) = \frac{2}{T} \int_0^{S_0} \frac{(K-x)^+}{KS_0} dK + \frac{2}{T} \int_{S_0}^{\infty} \frac{(x-K)^+}{KS_0} dK$$

Thus, the variance swap can be replicated with a portfolio of calls and puts of all strikes, weighted by $f_G''(K)$ plus a stochastic integral

$$-\frac{2}{TS_0} \int_0^T \log \frac{S_t}{S_0} dS_t.$$

Since the interest rate is zero, the price of the stochastic integral term is zero and the price of the option portfolio is

$$\sigma_K^2 = \frac{2}{T} \int_0^{S_0} \frac{P(T, K)}{KS_0} dK + \frac{2}{T} \int_{S_0}^{\infty} \frac{C(T, K)}{KS_0} dK$$

3. It is not possible to super-replicate the gamma swap with a finite number of options, because at maturity, the portfolio containing a finite number of options is ultimately linear in S_T , but the function f_G has superlinear growth as $S_T \rightarrow \infty$. However, since it is bounded by S_T^2 , it is possible to superreplicate the swap with a portfolio of options plus a quadratic pay-off.
4. This follows easily by the Black-Scholes formula.

5. The option prices are given by

$$P(T, K) = P^{BS}(T, K, I(T, K)), \quad C(T, K) = C^{BS}(T, K, I(T, K)).$$

Using (twice) the previous question and the symmetry property, we then have:

$$\begin{aligned} C(T, K) &= C^{BS}(T, K, I(T, K)) = \frac{K}{S_0} P^{BS}(T, S_0^2/K, I(T, K)) \\ &= \frac{K}{S_0} P^{BS}(T, S_0^2/K, I(T, S_0^2/K)) = \frac{K}{S_0} P(T, S_0^2/K). \end{aligned}$$

6. Combining questions 2 and 5, and making a change of variable $V = S_0^2/K$, we have:

$$\begin{aligned} \sigma_K^2 &= \frac{2}{T} \int_0^{S_0} \frac{P(T, K)}{K S_0} dK + \frac{2}{T} \int_{S_0}^{\infty} \frac{C(T, K)}{K S_0} dK \\ &= \frac{2}{T} \int_0^{S_0} \frac{C(T, S_0^2/K)}{S_0^2} dK + \frac{2}{T} \int_{S_0}^{\infty} \frac{P(T, S_0^2/K)}{S_0^2} dK \\ &= \frac{2}{T} \int_{S_0}^{\infty} \frac{C(T, V)}{V^2} dV + \frac{2}{T} \int_0^{S_0} \frac{P(T, V)}{V^2} dV \end{aligned}$$

Exercise 2

1.

$$dX_t = \pi_t \frac{dS_t}{S_t} + (X_t - \pi_t) r dt, \quad d\tilde{X}_t = \tilde{\pi}_t \sigma(\lambda dt + dW_t)$$

2. $Z_T = e^{-\lambda W_T - \frac{\lambda^2 T}{2}}$. Then, introducing, as in the course, $\tau_n = \inf\{t > 0 : \int_0^t \tilde{\pi}_s^2 ds \geq n\} \wedge T$, we have, by Fatou's lemma:

$$\begin{aligned} \mathbb{E}[Z_T \tilde{X}_T^{x, \pi}] &= \mathbb{E}^{\mathbb{Q}}[\tilde{X}_T^{x, \pi}] = \mathbb{E}^{\mathbb{Q}}[\lim_{n \rightarrow \infty} \tilde{X}_{\tau_n}^{x, \pi}] = \mathbb{E}^{\mathbb{Q}}[\lim_{n \rightarrow \infty} (\tilde{X}_{\tau_n}^{x, \pi} - M_{\tau_n})] + \mathbb{E}^{\mathbb{Q}}[M_0] \\ &\leq \lim_{n \rightarrow \infty} \mathbb{E}^{\mathbb{Q}}[(\tilde{X}_{\tau_n}^{x, \pi} - M_{\tau_n})] + \mathbb{E}^{\mathbb{Q}}[M_0] = x. \end{aligned}$$

3. By definition of V , for all u, v , $V(v) \geq U(u) - uv$. Taking $u = \tilde{X}_T^{x, \pi}$ and $v = yZ_T$, taking the expectation and using the previous question, we get the required result.

4. The computation of V is straightforward. Substituting this into the formula of the previous question, we have:

$$\mathbb{E}[U(\tilde{X}_T^{x, \pi})] \leq \frac{y}{\gamma} \log \frac{y}{\gamma} + \frac{y}{\gamma} \mathbb{E}[Z_T \log Z_T] - \frac{y}{\gamma} + xy$$

The 1st order condition for minimizing the RHS over y writes:

$$\frac{1}{\gamma} \log \frac{y}{\gamma} + \frac{1}{\gamma} \mathbb{E}[Z_T \log Z_T] + x = 0.$$

Solving for y and plugging this into the above formula yields

$$\mathbb{E}[U(\tilde{X}_T^{x, \pi})] \leq -e^{-\gamma x - \mathbb{E}[Z_T \log Z_T]}.$$

On the other hand,

$$\mathbb{E}[Z_T \log Z_T] = \mathbb{E}^{\mathbb{Q}} \left[-\lambda W_T - \frac{\lambda^2 T}{2} \right] = \frac{\lambda^2 T}{2}.$$

5. If $\lambda = 0$ then by previous question, for any admissible strategy π ,

$$\mathbb{E}[U(\tilde{X}_T^{x,\pi})] \leq -e^{-\gamma x}.$$

On the other hand, taking $\pi = 0$, we have that $\tilde{X}_T^{x,\pi} = x$ so that

$$\mathbb{E}[U(\tilde{X}_T^{x,\pi})] = -e^{-\gamma x}.$$

and $\pi = 0$ is the optimal strategy.

6. When $\tilde{\pi}_t$ is constant, $\tilde{X}_T^{x,\pi} = x + \tilde{\pi}\sigma(\lambda T + W_T)$, so that

$$\mathbb{E}[U(\tilde{X}_T^{x,\pi})] = -e^{-\gamma x - \gamma \tilde{\pi} \sigma \lambda T + \frac{1}{2} \gamma^2 \tilde{\pi}^2 \sigma^2 T}$$

7. Taking $\tilde{\pi} = \frac{\lambda}{\sigma\gamma}$, we get

$$\mathbb{E}[U(\tilde{X}_T^{x,\pi})] = -e^{-\gamma x - \frac{\lambda^2 T}{2}},$$

hence this strategy is optimal.

8. Substituting the explicit form of the utility function into the definition of the utility price, and using the fact that $\tilde{X}_T^{x,\pi} = x + \tilde{X}_T^{0,\pi}$, we have:

$$\min_{\pi \in \mathcal{A}} \mathbb{E}[e^{-\gamma(x - p(G) + \tilde{X}_T^{0,\pi} + \tilde{G})}] = \min_{\pi \in \mathcal{A}} \mathbb{E}[e^{-\gamma(x + \tilde{X}_T^{0,\pi})}].$$

Taking the constant factors out of the expectation, we see that x cancels out and $p(G)$ is given explicitly by

$$p(G) = -\frac{1}{\gamma} \log \frac{\min_{\pi \in \mathcal{A}} \mathbb{E}[e^{-\gamma(\tilde{X}_T^{0,\pi} + \tilde{G})}]}{\min_{\pi \in \mathcal{A}} \mathbb{E}[e^{-\gamma \tilde{X}_T^{0,\pi}}]}.$$

9. Let π_t^* be the optimal strategy without contingent claim, computed above. Then, for any admissible strategy π ,

$$\tilde{X}_t^{0,\pi} = \tilde{X}_t^{0,\pi^*} + \tilde{X}_t^{0,\pi-\pi^*} = \frac{\lambda}{\gamma}(\lambda t + W_t) + \tilde{X}_t^{0,\pi-\pi^*}.$$

Since the first term is a $\mathbb{Q}$ -martingale, the strategy $\pi - \pi^*$ is also admissible. Therefore,

$$\begin{aligned} \min_{\pi \in \mathcal{A}} \mathbb{E}[e^{-\gamma(\tilde{X}_T^{0,\pi} + \tilde{G})}] &= \min_{\pi \in \mathcal{A}} \mathbb{E}[e^{-\gamma(\tilde{X}_T^{0,\pi^*} + \tilde{X}_T^{0,\pi-\pi^*} + \tilde{G})}] \\ &= \min_{\pi \in \mathcal{A}} \mathbb{E}[e^{-\lambda^2 T - \lambda W_T} e^{-\gamma(\tilde{X}_t^{0,\pi-\pi^*} + \tilde{G})}] = e^{-\frac{\lambda^2 T}{2}} \min_{\pi \in \mathcal{A}} \mathbb{E}^{\mathbb{Q}}[e^{-\gamma(\tilde{X}_t^{0,\pi-\pi^*} + \tilde{G})}] \end{aligned}$$

On the other hand, we have seen that $\min_{\pi \in \mathcal{A}} \mathbb{E}[e^{-\gamma \tilde{X}_T^{0,\pi}}] = e^{-\frac{\lambda^2 T}{2}}.$

10. The monotonicity of the indifference price follows from the monotonicity of the utility function and the logarithm. To show convexity, consider two contingent claims G_1 and G_2 and two admissible strategies π_1 and π_2 . Then, by Cauchy-Schwarz inequality

$$\frac{1}{2} \log \mathbb{E}^{\mathbb{Q}}[e^{-\gamma(\tilde{X}_T^{0,\pi_1} + G_1)}] + \frac{1}{2} \log \mathbb{E}^{\mathbb{Q}}[e^{-\gamma(\tilde{X}_T^{0,\pi_2} + G_2)}] \geq \log \mathbb{E}^{\mathbb{Q}}[e^{-\frac{\gamma}{2}(\tilde{X}_T^{0,\pi_1+\pi_2} + G_1 + G_2)}]$$

Minimizing both sides over π_1 and π_2 yields the desired inequality. Then, by convexity,

$$p(G) - p'(G) = p(G) + p(-G) \leq 2p(0) = 0.$$

11. Let $\pi \in \mathcal{A}$ be an admissible strategy and let $x_G = \mathbb{E}^{\mathbb{Q}}[\tilde{G}]$. Since the replicating portfolio for G satisfies $\tilde{G}_t^{x_G, \pi_G} = \mathbb{E}^{\mathbb{Q}}[\tilde{G} | \mathcal{F}_t]$, it is a martingale and therefore the strategy $\pi + \pi_G$ is admissible. Then,

$$\begin{aligned} p(G) &= -\frac{1}{\gamma} \log \min_{\pi \in \mathcal{A}} \mathbb{E}^{\mathbb{Q}}[e^{-\gamma(\tilde{X}_T^{0,\pi-\pi_G} + x_G - \tilde{X}_T^{x_G, \pi_G} + \tilde{G})}] \\ &= x_G - \frac{1}{\gamma} \log \min_{\pi \in \mathcal{A}} \mathbb{E}^{\mathbb{Q}}[e^{-\gamma \tilde{X}_T^{0,\pi}}] = x_G, \end{aligned}$$

where the last equality follows from question 5.